06 - Text + Image Fusion

Goal: combine the text and image models and check if this beats text alone.

1. Turn the text SVM scores into probabilities.
2. Match text and image probabilities on the same 16984 validation products.
3. Compare three ways to combine them:
    - **Simple fusion:** a weighted average, with one fixed weight for text and image.
    - **Random Forest** (bagging): many decision trees trained on random samples, which then vote.
    - **Gradient Boosting** (boosting): trees trained one after another, each one fixing the errors of the previous ones.

Random Forest and Gradient Boosting learn how to combine the two models. Their input is, for each product, the 27 text probabilities and the 27 image probabilities (54 numbers).

Finally, we add the CamemBERT text model (notebook 05) as a third input.

1. Text probabilities

We use the optimised text model from notebook 04 (all features, binary TF-IDF, LinearSVC with C = 1). A LinearSVC gives scores, not probabilities. Calibration learns on the training set (3-fold CV) how to turn these scores into probabilities.

```
Text probabilities saved: (16984, 28)
Each row sums to 1: True
```

2. Match text and image on the same products

```
Products scored by both: 16984
```

3. Compare the three ways to combine

Text and image probabilities exist for the 16984 validation products, which neither model was trained on. We score every method with 5-fold cross-validation on these products: the combiner learns on 4/5 of them and predicts the remaining 1/5. Every product is predicted by a combiner that never saw it.

For simple fusion, the text weight (0 to 1, in steps of 0.1) is chosen on the 4/5 part only, in the same way.

```
Text weight chosen in each fold: [0.7, 0.7, 0.7, 0.7, 0.7]
```

Model	Weighted F1	Macro F1	Training time (s)
Image only	0.4817	0.4230	-
Text only	0.8430	0.8326	-
Simple fusion	0.8501	0.8401	-
Random Forest	0.8630	0.8491	33
Gradient Boosting	0.8595	0.8454	228

Analysis

- Simple fusion: the text weight chosen in every fold is 0.7 (70% text, 30% image). Weighted F1 0.8501, against 0.8430 for text only.
- Both learned methods beat simple fusion.
- **Random Forest is best:** weighted F1 0.8630, which is +1.3 points over simple fusion and +2.0 points over text only.
- Gradient Boosting is close (0.8595) but about 6 times slower.

4. Per category: best method vs text only

```
Best method: Random Forest
```

Code	Category	Text F1	Random Forest F1	Gain
2705	New Books	0.711	0.850	0.139
10	Used Books	0.566	0.663	0.097
2403	Book Lots	0.768	0.819	0.051
2280	Magazines	0.810	0.853	0.043
1280	Toys	0.712	0.747	0.035
40	Games	0.753	0.785	0.032
2220	Pets	0.883	0.906	0.023
1300	RC Models	0.926	0.945	0.018
2582	Garden Furniture	0.768	0.786	0.018
1160	Cards	0.954	0.971	0.017
2060	Decor	0.828	0.837	0.009
60	Consoles	0.879	0.887	0.008
1140	Figurines	0.830	0.838	0.008
2585	Garden Tools	0.833	0.839	0.006
1281	Board Games	0.616	0.622	0.006
2462	Used Games	0.825	0.829	0.005
1302	Outdoor Toys	0.837	0.841	0.004
1320	Baby	0.862	0.865	0.003
2522	Stationery	0.944	0.947	0.003
1560	Furniture	0.852	0.852	0.000
2583	Pool	0.981	0.981	0.000
2905	PC Games	0.988	0.988	0.000
1920	Textiles	0.920	0.920	-0.000
1301	Baby Socks	0.972	0.972	-0.000
50	Game Accessories	0.836	0.829	-0.007
1940	Food	0.927	0.911	-0.016
1180	Wargames	0.697	0.640	-0.057

Analysis

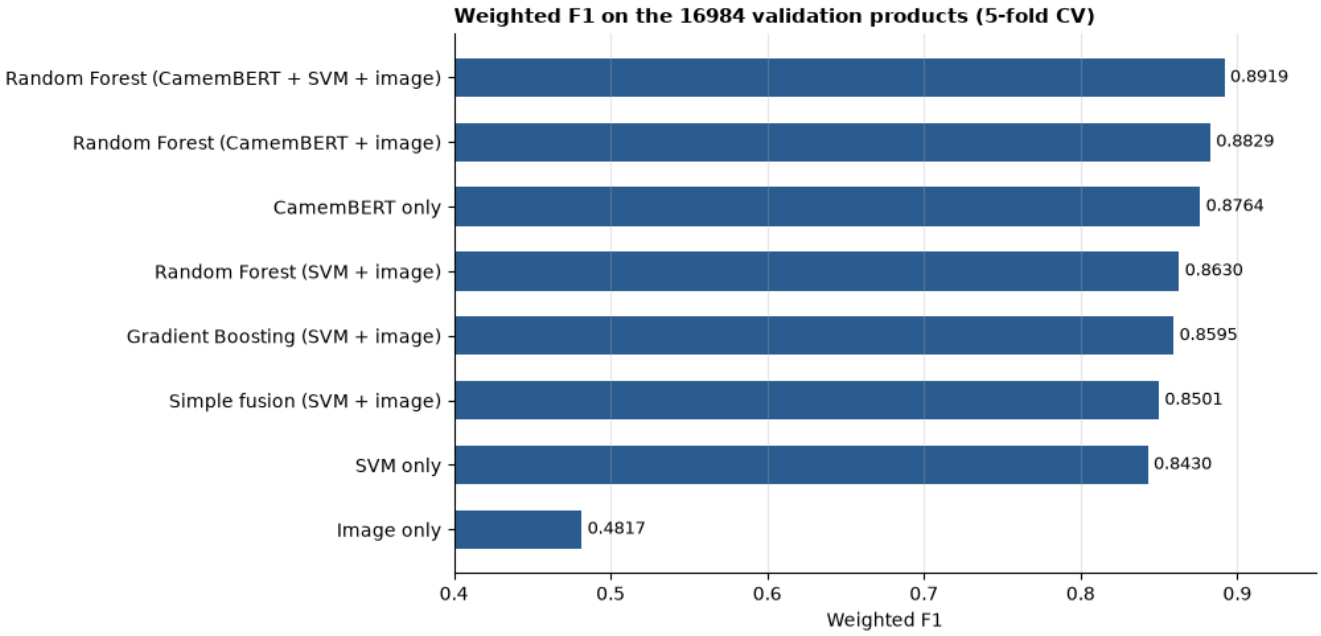
- The book categories gain the most: New Books +13.9 points, Used Books +9.7, Book Lots +5.1, Magazines +4.3.
- Random Forest learned to trust the image more for books. A single fixed weight cannot do this.
- A few small categories lose: Wargames -5.7 points and Food -1.6 (153 and 161 products). With few examples, the combiner has little to learn from.

5. Add CamemBERT

CamemBERT (notebook 05) is a stronger text model than the SVM. We give the Random Forest its 27 probabilities too, and test two inputs:

- CamemBERT + image (54 numbers)
- CamemBERT + SVM + image (81 numbers)

Same 5-fold cross-validation as above.



Model	Weighted F1	Macro F1
Random Forest (CamemBERT + SVM + image)	0.8919	0.8785
Random Forest (CamemBERT + image)	0.8829	0.8682
CamemBERT only	0.8764	0.8605
Random Forest (SVM + image)	0.8630	0.8491
Gradient Boosting (SVM + image)	0.8595	0.8454
Simple fusion (SVM + image)	0.8501	0.8401
SVM only	0.8430	0.8326
Image only	0.4817	0.4230

Analysis

- Adding CamemBERT gives the best results.
- **Best model: Random Forest on CamemBERT + SVM + image, with 0.8919 weighted F1** and 0.8785 macro F1.
- That is +1.5 points over CamemBERT alone (0.8764) and +4.9 points over the SVM alone (0.8430).
- The SVM still adds value next to CamemBERT (0.8919 against 0.8829 without it). The two text models make different errors: CamemBERT reads context, the SVM catches rare specific words.
- The image adds about 0.7 points on top of CamemBERT alone (0.8829 against 0.8764). It helps less than with the SVM, because CamemBERT already fixes many book errors.

Conclusion

- Combining models always helps, and a learned combination (Random Forest) works better than a fixed weight.
- Best model: Random Forest on the CamemBERT, SVM and image probabilities, with 0.8919 weighted F1 on the validation products (5-fold CV).
- Each part adds something: CamemBERT gives the best text understanding, the SVM catches rare specific words, and the image helps on books.